Date: march 22, 2026 (Sun) ME302: Energy Systems – II Due date: april 11, 2026 (Sat)

Term: 2025–26, Semester II

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Assignment–II

Each question carries an equal weightage.

Question 1

A multi-stage high-pressure steam turbine is supplied with steam at a stagnation pressure of 7 MPa (abs.) and a stagnation temperature of $500°C$. The corresponding specific enthalpy is 3410 kJ/kg.

The steam exhausts from the turbine at a stagnation pressure of 0.7 MPa (abs.), the steam having been in a superheated condition throughout the expansion. It can be assumed that the steam behaves like a perfect gas over the range of expansion and that $\gamma = 1.3$.

Given that the turbine flow process has a small-stage efficiency of 0.82, determine:

1. the temperature and specific volume at the end of the expansion,
2. the reheat factor.

The specific volume of superheated steam is represented by pv = 0.231(h - 1943) where p is in kPa, v is in m^3/kg, and h is in kJ/kg.

Question 2

The air entering the impeller of a centrifugal compressor has an absolute axial velocity of 100 m/s. At rotor exit, the relative air angle measured from the radial direction is $26°36'$. The radial component of velocity at exit is 120 m/s, and the tip speed of the radial vanes is 500 m/s.

Determine the power required to drive the compressor when the air flow rate is 2.5 kg/s and the mechanical efficiency is 95%.

If the radius ratio of the impeller eye is 0.3, calculate a suitable inlet diameter assuming that the inlet flow is incompressible.

Determine the overall total pressure ratio of the compressor when the total-to-total efficiency is 80%, assuming that the velocity at exit from the diffuser is negligible.

Question 3

Values of pressure (kPa) measured at various stations of a zero-reaction gas turbine stage, all at the mean blade height, are given below:

Stagnation Pressure (kPa)	**Static Pressure (kPa)**
Nozzle entry: 414	Nozzle exit: 207
Nozzle exit: 400	Rotor exit: 200

The mean blade speed is 291 m/s, inlet stagnation temperature is 1100 K, and the flow angle at nozzle exit is 70° measured from the axial direction.

Assuming that the magnitude and direction of the velocities at entry and exit of the stage are the same, determine the total-to-total efficiency of the stage.

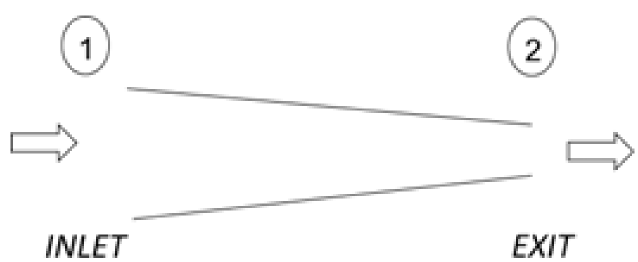
Assume a perfect gas with $C_p = 1.148\,\mathrm{kJ/(kg\,K)}$ and $\gamma = 1.333$.

Question 4

A conical converging nozzle, having an inlet diameter of 0.3 m and length 1 m, is to be designed to operate at choked exit condition, $M_2 = 1$. At inlet, the stagnation pressure is 225 kPa, stagnation temperature is 600 K and the Mach number is $M_1 = 0.5$.

The exit is exposed to the ambient having atmospheric pressure, 101.3 kPa. The fluid, air, can be considered an ideal gas with $\gamma = 1.4$ and $R = 0.287\,\mathrm{kJ/(kg\,K)}$. Assuming that the nozzle flow is adiabatic, answer the following:

1. For the ideal isentropic case with no losses, determine the nozzle exit diameter. Will the flow be ideally expanded, under-expanded or over-expanded?
2. If the stagnation pressure loss is equal to 2% of the inlet stagnation pressure, determine the nozzle exit diameter.



Question 5

Calculate the specific thrust of a simple stationary jet propulsion unit with the following data:

- Total head isentropic efficiency of compressor: 80%
- Total head isentropic efficiency of turbine: 85%
- Total head pressure ratio (including combustion chamber pressure loss): 4 : 1
- Combustion efficiency: 98%

- Mechanical transmission efficiency: 99%
- Maximum cycle temperature: 1000 K
- Air mass flow rate: 20 kg/s

For air: $C_p = 0.24\,\mathrm{kcal/(kg\,K)}$, $\gamma = 1.4$
For gases: $C_p = 0.276\,\mathrm{kcal/(kg\,K)}$, $\gamma = 1.33$

Ambient temperature and pressure are $15°C$ and $1\,\mathrm{kg/cm^2}$ respectively. Assume nozzle efficiency = 90%